

The maximum length of a chess game under the FIDE Laws is 17,697 plies

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Abstract

Under the FIDE Laws of Chess, which end a game automatically on a fivefold repetition or when 75 moves by each player pass without a pawn move or a capture, the length of a chess game is bounded. Labelle’s analysis [1], begun in 2015, identified the value 17,697 plies ($8,848\frac{1}{2}$ moves) by counting the clock-resetting moves available, gave three-switch schedules and a complete generated game, and left maximality a conjecture; Murphy [2] later published a separate complete construction of the same length, with a counting ceiling of 17,699. Neither source excludes 17,698 or 17,699. We prove that the maximum is exactly 17,697. Writing K for the number of critical segments of a game and S for the number of times the colour making the segment-ending moves changes, every game satisfies $L \leq 150K - S$; we show $K \leq 118$, and that $K = 118$ forces $S \geq 3$, which yields $L \leq 150 \cdot 118 - 3 = 17,697$. The argument is elementary, and every finite case analysis it appeals to is machine-checked in a companion repository, which also verifies the 17,697-ply witness move by move and re-decides the central case analysis by two independent mechanical methods.

1 Introduction

How long can a game of chess be? Under earlier versions of the Laws the question had no finite answer: the draw by threefold repetition and the 50-move rule must be *claimed* by a player, so two cooperating players could continue indefinitely. This changed on 1 July 2014, when FIDE introduced two automatic termination rules [3], retained in the current Laws [5]: the game is drawn, without any claim, when the same position occurs five times (Article 9.6.1), or when 75 consecutive moves by each player have been made without the movement of any pawn and without any capture (Article 9.6.2) — with the proviso that a checkmate delivered by the final move takes precedence. Under these Laws the supremum of game lengths is finite, and computing it exactly becomes a well-posed combinatorial problem.

The question has a substantial informal history, and the value itself is not new. For the older, claim-based 50-move setting — under the assumption that the claimable draws are in fact invoked — Labelle [1] computed a maximum of $5,898\frac{1}{2}$ moves. For the modern automatic rules, the same analysis, maintained since 2015, counts the clock-resetting moves

available (96 pawn moves and 30 captures, less 8 moves that must be both), presents explicit schedules — due in part to contributors of the forum threads the page credits — realising all of them at the cost of three half-moves, and states, as a belief supported by those constructions, that the maximum length is $118 \times 75.0 - 3 \times 0.5 = 8,848\frac{1}{2}$ moves; the page also supplies a complete generated game of that length. Murphy [2] subsequently published a separate, hand-designed complete game of 17,697 plies ($8,848\frac{1}{2}$ moves, ending in checkmate on White’s 8,849th move) — the witness verified in Section 6 — treating the exact maximum as open: his counting ceiling admits 17,699. Tromp’s reference note [6] records the value. Both sources thus establish the lower bound 17,697; neither excludes 17,698 or 17,699, and to our knowledge no proof of optimality has been published. This paper supplies one:

Theorem 1.1 (Main theorem, proved as Theorem 5.1). *No legal chess game lasts more than 17,697 plies, and the bound is attained: the maximum length of a chess game under the FIDE Laws is exactly 17,697 plies.*

Relative to the counting sketch of [1], the content of the theorem is twofold. First, the count of 118 is turned into an upper bound over *all* games: a game need not promote every pawn, so the deduction of eight overlapping moves cannot be read off the intended construction and must be derived — Proposition 3.6 couples the pawn-move and overlap counts through the number of resolved files — and the thirtieth capture interacts with the end of the game (Lemma 3.7). Second, and this is the bulk of the paper, the three half-move losses are proved unavoidable: no ordering of the critical moves of any 118-segment game achieves fewer than three colour changes (Theorem 4.6).

The shape of the proof. A *critical move* — a pawn move or a capture — resets the 75-move counter and is irreversible. Critical moves cut a game into *segments*, each holding at most 149 reversible plies plus the critical move that ends it, so a game with K segments has at most $150K$ plies; a parity argument charges one further ply each time the *colour* making the segment-ending moves changes, giving $L \leq 150K - S$ (Lemma 2.4). Counting pawn moves and captures yields $K \leq 118$ (Theorem 3.8), and $150 \cdot 118 - 1 = 17,699$ is the ceiling of [2]. The remaining two plies are removed in Section 4: a game attaining $K = 118$ is so constrained — all sixteen pawns must make six single-square moves each and promote, and at least 29 captures must occur — that its critical moves cannot be scheduled with fewer than three colour changes (Theorem 4.6). Four of the five candidate schedules are eliminated by counting alone; the fifth is eliminated by a lemma about pawns: *while the enemy’s pawns all stand on their home rank, an advancing pawn stops four moves short of promotion* (Lemma 4.4). Since $150 \cdot 117 = 17,550 < 17,697$, games with $K \leq 117$ are shorter still, and the theorem follows.

Nothing in the bound depends on how the game ends, which is why one argument covers checkmate, the 75-move rule, fivefold repetition, stalemate and dead positions alike.

Machine assistance, and what must be trusted. The proof is elementary and short enough to check by hand. Machine assistance enters at clearly delimited points, all described in Section 7: a handful of finite case analyses are exhausted mechanically (each also proved by hand in the text); the 17,697-ply witness is verified ply by ply; and the central case

analysis of Section 4 is re-decided by two independent mechanical methods, one a constraint solver and one solver-free, as a cross-check on the human argument. A companion repository [9] contains all of this, with a claims-to-code map reproduced in Appendix A. The trusted base is stated explicitly in Section 7.3; it consists of the FIDE Laws as summarised in Section 2 and, for the mechanical checks only, the move generator of the `python-chess` library [7].

2 Setting, critical moves, and the segment decomposition

2.1 The rules assumed

A *game* is a sequence of legal moves from the standard initial position, played until a rule of the FIDE Laws [5] ends it. We write L for its length in *plies* (single moves by one player) and number plies from 1; White plays the odd plies. The game-ending rules relevant here are:

1. *Checkmate* (Art. 5.1.1) and *stalemate* (Art. 5.2.1) end the game on the move that produces them.
2. *Dead position* (Art. 5.2.2): the game is drawn the moment a position arises from which no sequence of legal moves can lead to a checkmate by either side.
3. *Fifefold repetition* (Art. 9.6.1): the game is drawn, with no claim required, when the same position (same pieces on the same squares, same side to move, same castling and en passant possibilities, as in Art. 9.2.2) has occurred five times.¹
4. *Seventy-five-move rule* (Art. 9.6.2): the game is drawn, with no claim required, when at least 75 consecutive moves by each player — that is, 150 consecutive plies — have been played without any pawn move and without any capture. If the ply completing the 150 delivers checkmate, the checkmate stands.

Draws by agreement, resignations, and the *claimable* draws (threefold repetition, Art. 9.2; the 50-move rule, Art. 9.3) are voluntary: players maximising the length of a game simply never invoke them, so they play no role in the maximisation. This is the setting of [2].

It is convenient to track the *halfmove clock* $c(t)$: the number of plies played since the most recent pawn move or capture (or since the start of the game, if there has been none). Article 9.6.2 says precisely that the game ends when c reaches 150, the ply that brings it there being the last of the game, and being scored as mate if it mates.

2.2 Critical moves and segments

Definition 2.1 (Critical move). A *critical move* is a pawn move or a capture. (En passant is both; a promotion is a pawn move; castling is neither.) A *quiet* move is a move that is not critical.

¹The 2014 formulation required the five occurrences to be consecutive; the requirement was removed in the 2017 revision [4]. The difference is immaterial here: the upper bound follows from the automatic 75-move rule alone, and the verified witness never lets any position occur even three times (Section 6).

Critical moves are exactly the moves that reset the halfmove clock, and they are irreversible: a pawn cannot retreat, and a captured man never returns.

Definition 2.2 (Segments, endpoints, K , T). Let $t_1 < t_2 < \dots < t_m$ be the plies of a game's critical moves. Set $T = 1$ if quiet plies follow the last critical move (or if the game has moves but no critical move at all), and $T = 0$ otherwise. The game's *endpoints* are

$$e_i = t_i \ (1 \leq i \leq m), \quad e_{m+1} = L \text{ if } T = 1,$$

together with the virtual endpoint $e_0 = 0$. The number of *segments* is $K = m + T$; segment i consists of plies $e_{i-1} + 1, \dots, e_i$. Every ply lies in exactly one segment, and $e_K = L$.

Definition 2.3 (Switches S). The *colour* of an endpoint e_i ($i \geq 1$) is the colour of the player who played ply e_i — White if e_i is odd, Black if even. By convention the virtual endpoint $e_0 = 0$ is *Black* (it has the parity of a Black ply, and the game opens as if Black had just made a critical move). The switch count is

$$S = \#\{i \in \{1, \dots, K\} : \text{colour}(e_i) \neq \text{colour}(e_{i-1})\}.$$

Lemma 2.4 (Segment decomposition). *Every game satisfies*

$$L = 150K - S - \sum_{i=1}^K \delta_i \quad \text{with every } \delta_i \geq 0,$$

where δ_i is the shortfall of segment i from its target length (150 if endpoints e_{i-1}, e_i have the same colour, 149 otherwise). In particular $L \leq 150K - S$.

Proof. Fix i and consider the segment plies $e_{i-1} + 1, \dots, e_i$. All of them except possibly the last are quiet (t_i is the next critical ply after t_{i-1} ; the terminal segment is entirely quiet). Suppose $e_i - e_{i-1} \geq 151$. Then the 150 plies $e_{i-1} + 1, \dots, e_{i-1} + 150$ are all quiet, so after ply $e_{i-1} + 150$ the halfmove clock reads 150 and the game has ended there by Article 9.6.2 (as a draw, or as the mate that takes precedence — either way it has ended), contradicting the existence of ply $e_{i-1} + 151 \leq e_i \leq L$. Hence $e_i - e_{i-1} \leq 150$.

Plies alternate colours, so $e_i - e_{i-1}$ is even exactly when e_i and e_{i-1} have the same colour; this is where the convention $\text{colour}(e_0) = \text{Black}$ is used, 0 having the parity of a Black ply. A segment whose endpoints differ in colour therefore has odd length, and being ≤ 150 , has length ≤ 149 . Summing the segment lengths telescopes to $e_K - e_0 = L$ and yields the identity. $\square$

Remark 2.5. The terminal quiet segment genuinely can reach 150 plies: if the game ends by the 75-move rule, the drawing ply is the 150th quiet ply after the last critical move and it *is played*. Its parity then forces the terminal endpoint to have the same colour as the previous one, consistent with the accounting above. A terminal segment ending earlier — by a quiet checkmate, a stalemate, a fivefold repetition or a dead position — is only shorter.

Counting critical moves by inclusion–exclusion, with P the number of pawn moves, C the number of captures and O the number of *overlaps* (moves that are both, i.e. diagonal pawn captures, en passant included),

$$K = (P - O) + (C + T). \tag{1}$$

The next two sections bound the two halves of (1) separately. Neither half refers to how the game ends.

3 At most 118 critical segments

Throughout, we count over the 32 *units* that begin the game — each man followed through the whole game under promotion — rather than over piece types. A unit that begins as a pawn is still that unit after promoting.

Lemma 3.1 (Pawn basics). *Each pawn makes at most 6 pawn moves, so $P \leq 96$. A pawn that makes exactly 6 pawn moves makes only single-square moves and promotes.*

Proof. Every pawn move advances the pawn at least one rank toward promotion, and there are six ranks between its home rank and the promotion rank. A pawn making six moves therefore advances exactly one rank per move and finishes on the promotion rank, where promotion is mandatory. $\square$

3.1 Origin pairs and the bound $P - O \leq 88$

Definition 3.2 (Origin pair, resolved file). The two pawns that begin on file i (one White, one Black) are that file’s *origin pair*. File i is *resolved* if at least one of those two pawns makes a diagonal pawn move at some point in the game; otherwise it is *unresolved*. Write f for the number of resolved files.

A diagonal pawn move is the only way a pawn changes file, and it is always a capture; so both pawns of an unresolved pair spend their entire lives on file i .

Lemma 3.3 (Unresolved pair cap). *The two pawns of an unresolved origin pair make at most 10 pawn moves combined, and 10 is attained in a legal game.*

Proof. Both pawns stay on file i . While both are on the board, the White pawn is strictly below the Black one: neither can capture the other (a capture would be diagonal), and neither can pass the other, since a single push onto an occupied square is illegal and a double push may not jump over an occupied square. So if White’s pawn has advanced a ranks and Black’s b ranks, they stand on ranks $2 + a < 7 - b$, giving $a + b \leq 4$; and since each pawn move advances at least one rank, the two pawns have made at most 4 combined moves while both are on the board. In particular neither pawn can promote while the other is on the board, so at most one of the pair ever promotes, and only after the other has been captured (necessarily by some other unit).

When one pawn of the pair leaves the board its total contribution is frozen at some $d \leq 4$, and the survivor’s lifetime total is at most 6 by Lemma 3.1, for a combined total of at most $d + 6 \leq 10$.

Ten is attained: in

1. Nf3 a6 2. Ng1 a5 3. Nf3 a4 4. Ng1 a3 5. Nxa3 Nc6 6. Nc4 Rb8 7. a3 Nf6 8. a4 Ng8
9. a5 Nf6 10. a6 Ng8 11. a7 Nf6 12. a8=Q Ng8

Black’s a-pawn walks $a7$ – $a3$ in four moves and is captured there by the knight; White’s a-pawn then walks the whole file and promotes, six more. Neither pawn ever moved diagonally, so the a-file is unresolved throughout. The sequence is a legal prefix rather than a completed

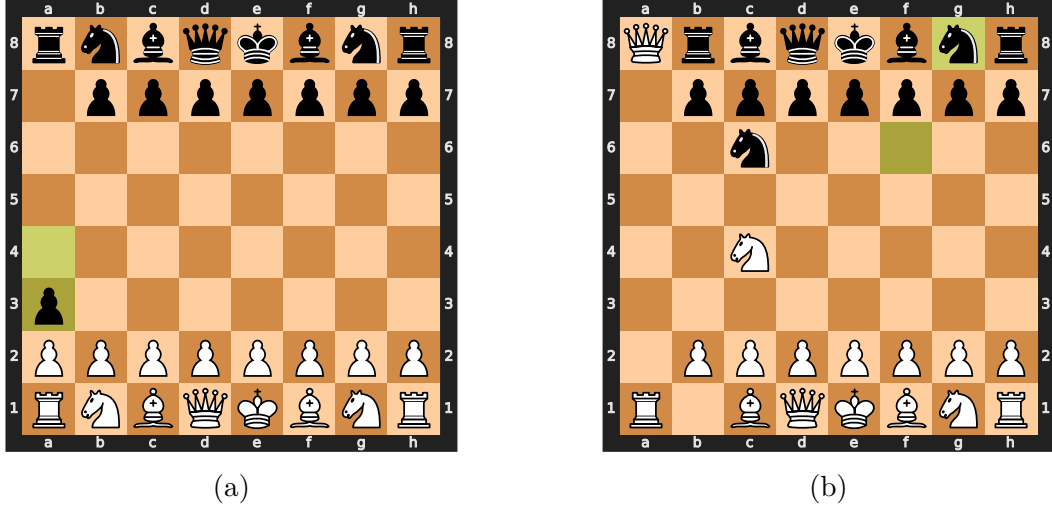


Figure 1: The unresolved origin pair attaining ten combined pawn moves (Lemma 3.3). (a) After 4...a3: Black's a-pawn has made four moves and is captured on the next move by the b1-knight. (b) The final position: White's a-pawn has walked the file and promoted, six further moves. Neither pawn ever left the a-file.

game, but in any legal completion neither count changes — the survivor has promoted and the other has been captured, so the pair makes no further pawn move and the file stays unresolved — so a completed legal game attains ten as well. Figure 1 shows the position after Black's fourth pawn move and the final position of the sequence. $\square$

Remark 3.4. Both halves of the argument are additionally re-derived mechanically, by exhausting a state space strictly more permissive than chess (turn order ignored, no third unit blocking the file, either pawn removable at any moment without cost); see Appendix A. We record a caution for anyone reproducing this bound: the first half of the argument alone — “two facing pawns cannot pass, so at most 4” — is *false* as a cap, since it ignores that a captured pawn frees the survivor. A constraint stronger than legality produces spurious impossibility results; see Appendix B.

Lemma 3.5 (Overlap floor). $O \geq f$.

Proof. For each resolved file i fix a witnessing diagonal pawn move; it is a capture by a pawn whose *origin* file is i . A pawn has one origin file, so the witnessing moves of distinct resolved files have distinct movers and are distinct moves, each of them an overlap. $\square$

Proposition 3.6. $P - O \leq 88$, and equality forces $f = 8$, $P = 96$ and $O = 8$.

Proof. The eight origin pairs partition the sixteen pawns, so summing Lemma 3.3 over unresolved pairs and the trivial cap 12 over resolved pairs, and combining with Lemma 3.1,

$$P \leq \min(96, 10(8 - f) + 12f) = \min(96, 80 + 2f).$$

With Lemma 3.5, $P - O \leq \min(96, 80 + 2f) - f$. For $f \leq 7$ this is at most $80 + 2f - f = 80 + f \leq 87$; for $f = 8$ it is $96 - 8 = 88$. The maximum over f is therefore 88, attained only at $f = 8$ with $P = 96$ and $O = f = 8$. $\square$

Maximising over f is what makes Proposition 3.6 a statement about all games rather than about games with eight overlaps; asserting $O \geq 8$ outright would be circular, being true only *because* $K = 118$ forces every file to be resolved.

3.2 The bound $C + T \leq 30$

Lemma 3.7. $C + T \leq 30$.

Proof. Only the 30 non-king units can ever be captured, so $C \leq 30$. If $C \leq 29$ then, since $T \leq 1$, we get $C + T \leq 30$. If $C = 30$, the 30th capture leaves the two kings alone on the board. King versus king is a dead position — no sequence of legal moves mates — so by Article 5.2.2 the game ends on that very ply; no quiet ply follows the last critical move and $T = 0$, giving $C + T = 30$. $\square$

This is the only place any dead-position reasoning enters the upper bound, and king versus king is the one case where deadness is beyond dispute. The two ways of reaching 30 are worth displaying, because they explain why draws do not outlast checkmates:

$$\underbrace{96 + 30 - 8 + 0}_{\text{all 30 captured, no closing segment}} = \underbrace{96 + 29 - 8 + 1}_{\text{29 captured, one closing segment}} = 118.$$

3.3 The bound $K \leq 118$ and its equality case

Theorem 3.8. *Every game has $K \leq 118$. Moreover a game with $K = 118$ satisfies*

$$f = 8, \quad P = 96, \quad O = 8, \quad C + T = 30,$$

so all sixteen pawns make six single-square pawn moves each and promote, and $C \geq 29$.

Proof. By (1), Proposition 3.6 and Lemma 3.7,

$$K = (P - O) + (C + T) \leq 88 + 30 = 118.$$

Equality forces $P - O = 88$ and $C + T = 30$; the former pins $(f, P, O) = (8, 96, 8)$ by the equality case of Proposition 3.6, and $P = 96$ makes every pawn's count exactly 6, whence Lemma 3.1 applies to each. Finally $T \leq 1$ gives $C \geq 29$. $\square$

Remark 3.9. The uniqueness of the equality assignment is also verified by mechanical enumeration of every $(f, P, O, C+T)$ the constraints admit at $K = 118$ (there is exactly one) and at $K = 117$ (there are many, which is what makes the pinning at 118 informative). See Appendix A.

Theorem 3.8 with $S \geq 1$ already gives Murphy's ceiling $150 \cdot 118 - 1 = 17,699$. The next section removes the last two plies.

4 Games with 118 segments have at least three switches

Throughout this section, fix a game with $K = 118$; by Theorem 3.8 it has $P = 96$ — *every pawn makes exactly six pawn moves* — and $C \geq 29$. These two facts do all the work. We emphasise at the outset that $S \geq 3$ is claimed *only* under $K = 118$; it is false in general (Remark 4.7).

4.1 Block shapes and the terminal seam

The critical moves of a game, in order, form a sequence of coloured events (the colour of a pawn move is the pawn's owner; the colour of a capture is the capturer, i.e. the side that does *not* own the victim). Group maximal runs of equal colour into *blocks*; the resulting alternating sequence of block colours is the game's *critical shape*. Counting the virtual Black endpoint, an alternating sequence of n blocks contributes $n - 1$ switches if it opens with Black and n if it opens with White.

The switch count S of Definition 2.3 is measured over all K endpoints, the last of which may be the terminal quiet endpoint rather than a critical move. The two viewpoints line up in one step:

Lemma 4.1 (Terminal seam). *The endpoint sequence of a game is its critical-move sequence with, when $T = 1$, one endpoint appended. Deleting a final element cannot increase the number of adjacent unequal pairs, so if a game has $S \leq 2$ then its critical shape has at most 2 switches as well.*

Proof. Immediate. (In the companion repository the claim is additionally exhausted over both readings of every alternating shape of up to twelve blocks.) $\square$

A game with $K = 118$ has at least 117 critical moves, so its critical shape is nonempty; and a nonempty alternating shape with at most 2 switches is one of exactly five:

$$B, \quad W, \quad BW, \quad WB, \quad BWB,$$

since n blocks contribute at least $n - 1$ switches. We refute all five. Each side has 15 capturable units, of which exactly 7 (queen, two rooks, two bishops, two knights) did not begin as pawns; the count below is over units, so a pawn-origin unit captured after promoting still counts against the pawn origins.

4.2 Four shapes are eliminated by counting

Proposition 4.2 (Single block). *Shapes B and W are impossible at $K = 118$.*

Proof. A pawn move is a critical move of the pawn's own colour. In a single-block game only one colour makes critical moves, so only its eight pawns ever move: $P \leq 48 < 96$. $\square$

Proposition 4.3 (Two blocks). *Shapes BW and WB are impossible at $K = 118$.*

Proof. Write the shape (X, Y) : every critical move of X precedes every critical move of Y . Since Y captures only X 's units, $C_Y \leq 15$, so $C_X \geq C - 15 \geq 29 - 15 = 14$: the colour moving first makes at least 14 captures, all of them before Y has made a single critical move. Each victim is a Y unit that, being dead before every Y critical move, never made a pawn move. At most 7 of the victims are non-pawn-origin units, so at least 7 pawn-origin Y units make no pawn move at all, and

$$P \leq 96 - 7 \cdot 6 = 54 < 96. \quad \square$$

Note that Y may well have made *quiet* moves in the meantime; the argument only needs its critical moves to come later.

4.3 The home-rank lemma

One shape is not eliminated by counting alone; the following lemma closes it.

Lemma 4.4 (Home-rank lemma). *Fix colours A (attacker) and D (defender), and consider any position reachable from the initial position by a sequence of moves satisfying*

(R1) *every move by D is quiet, and*

(R2) *no move by A captures a D pawn.*

Then every square of D 's pawn home rank still holds the D pawn that started there; no A pawn ever stands on that rank; and no A pawn has made more than four pawn moves.

Proof. We use the following facts about the rules, labelled for Section 7, where their mechanical treatment is described: [H0] the initial position has eight pawns on each home rank; [H1] a legal move starts on a square holding a piece of the side to move; [H2] a move onto an occupied square is a capture of an enemy piece standing there; [H3] castling changes only back-rank squares; [H4] an en passant capture removes a pawn from rank 4 or rank 5; [H5] a move changes the occupancy of no squares other than its origin and destination, the castling rook's two squares, and the en passant victim's square; [H6] a pawn move advances the pawn exactly one rank, or exactly two from its own home rank only.

Part 1: D 's home rank never changes. Induction on moves, the base case being [H0]. Let the invariant hold and let m be a move permitted by (R1), (R2). By [H5] it suffices to rule out each way m could touch a home-rank square. m cannot *start* there: the square holds a D pawn, only D may move it [H1], and that would be a pawn move, excluded by (R1). m cannot *end* there: the square is occupied, so the move would capture an enemy piece standing there [H2]; D capturing is excluded by (R1), and the occupant is a D pawn, so A capturing it is excluded by (R2). Castling touches only back ranks [H3]. En passant removes a pawn from rank 4 or 5 [H4], not from a home rank (and its victim would in any case have just made a double push — for a D victim a pawn move, excluded by (R1)).

Part 2: no A pawn reaches D 's home rank. Standing there requires a move ending there, ruled out above.

Part 3: at most four moves. By [H6] a pawn never moves backwards and double-pushes only from its own home rank. By Part 2 an A pawn never occupies D 's home rank; nor the rank beyond it, which could only be reached by a single step from D 's home rank or a double push from a rank that is not the pawn's home rank. Its range is therefore the four ranks between its own home rank and D 's, and each pawn move spends at least one of them: at most four moves. (Four is attained, by four single pushes.) $\square$

Figure 2 shows a minimal instance of the confinement.

Proposition 4.5 (Three blocks). *Shape BWB is impossible at $K = 118$.*

Proof. All of White's critical moves lie in the middle block. Let τ be the ply of White's first critical move, and call the plies before τ the *first phase*. As in Proposition 4.3, Black captures at most 15 units, so White makes $C_W \geq 14$ captures, all in the middle block; at least $14 - 7 = 7$ of the victims are pawn-origin Black units.

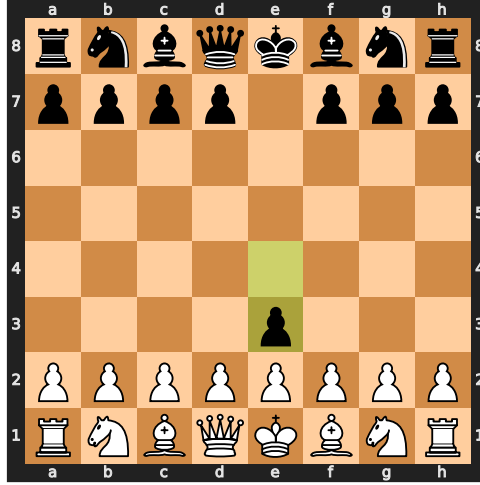


Figure 2: The home-rank lemma (Lemma 4.4), minimal instance. After 1. Nf3 e6 2. Ng1 e5 3. Nf3 e4 4. Ng1 e3, Black’s pawn has made four moves and is confined: the square e2 holds a pawn that only White may move, which would violate (R1), and the captures on d2 and f2 are exactly what (R2) forbids.

Fix such a victim. By Theorem 3.8 it makes exactly six pawn moves, all before its capture. Its pawn moves are Black critical moves, and Black critical moves at plies $> \tau$ belong to the third block, hence come after *every* White critical move — in particular after the capture, which is impossible for a dead unit. So all six of its pawn moves occur in the first phase.

Now apply Lemma 4.4 to the first phase with $A = \text{Black}$, $D = \text{White}$. (R1) holds by the definition of τ : every White move before τ is quiet. (R2) holds because a White pawn captured before τ would be dead before every White critical move and so would make none of its six pawn moves, contradicting $P = 96$. The lemma caps every Black pawn at four pawn moves in the first phase — but each of the seven victims needs six there. Even charging only the difference,

$$P \leq 96 - 7 \cdot 2 = 82 < 96. \quad \square$$

The only escape the home-rank lemma allows — capturing a way through the home rank — costs the captured pawn all six of its moves, which Theorem 3.8 cannot spare. That is the entire content of clause (R2) above.

Theorem 4.6. *Every game with $K = 118$ has $S \geq 3$.*

Proof. Suppose $S \leq 2$. By Lemma 4.1 the game’s critical shape has at most two switches, so it is one of the five shapes listed above; each is refuted by Proposition 4.2, 4.3 or 4.5. $\square$

Remark 4.7. $S \geq 3$ is conditional on $K = 118$, and only on it: every step above runs on $P = 96$ and $C \geq 29$. It is plainly false in general. The 16-ply game 1. Nf3 Nf6 2. Ng1 Ng8 repeated four times is legal and ends in fivefold repetition of the initial position; it has no pawn move and no capture, so $K = 1$ (the single closing segment), and its one endpoint is

a Black ply, matching the virtual Black opening: $S = 0$. This costs nothing in Section 5: a game with $K \leq 117$ forfeits at least 150 plies, more than any switch count could recover.

5 The main theorem

Theorem 5.1. *No legal chess game exceeds 17,697 plies, and the game constructed in [2] attains this length. The longest chess game is exactly 17,697 plies — 8,848 full moves and one final White ply, delivering checkmate on White’s 8,849th move.*

Proof. Let a game have K segments. If $K \leq 117$ then by Lemma 2.4 with the trivial $S \geq 0$,

$$L \leq 150 \cdot 117 = 17,550 < 17,697.$$

If $K = 118$ — the largest value Theorem 3.8 permits — then Theorem 4.6 gives $S \geq 3$, so

$$L \leq 150 \cdot 118 - 3 = 17,697.$$

The bound is attained: Murphy’s game is legal and 17,697 plies long, which we verify mechanically, ply by ply, in Section 6. $\square$

The two cases must be kept separate, because $S \geq 3$ is not available at $K \leq 117$ (Remark 4.7); it is also not needed there, since one whole segment outweighs any possible switch count.

6 The witness

Among the published maximum-length games we verify the one constructed by hand in [2]; Labelle’s page [1] supplies another, machine-generated. The companion repository [9] contains an independent verifier for it (built on the move generation of `python-chess` [7] and its own implementation of the termination rules of Section 2.1), together with a per-ply trace. The verified facts:

- The game is legal, 17,697 plies long, and ends in checkmate, in the final position `3k4/3Q4/4K3/8/8/8/8/8` — king and queen against king, Figure 3.
- No automatic termination fires early. The halfmove clock stays below 150 after every ply but the last; no position ever occurs even three times (the maximum multiplicity of any position over the whole game, under the identity of Article 9.2.2, is *two*), so fivefold repetition is never approached; no stalemate occurs.
- No position of the game is dead, by an argument that needs no material test: from every position, the remaining moves of the game are themselves a legal series ending in checkmate, and a position from which checkmate is reachable is precisely what Article 5.2.2 does not end. The verifier checks the legality of those suffixes; its material-based deadness test is not load-bearing here (see Section 7.3).

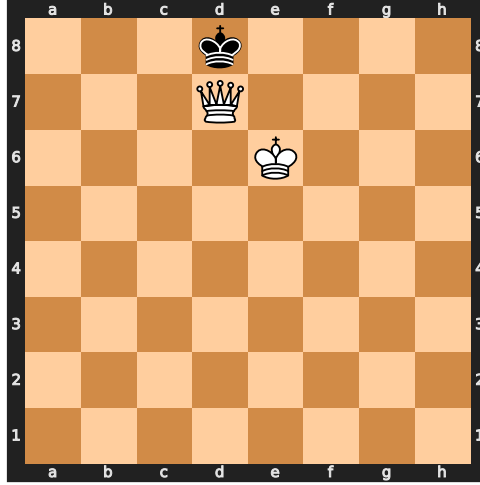


Figure 3: The final position of the witness, after ply 17,697: checkmate, delivered with the halfmove clock at exactly 150.

- The final ply is a quiet checkmate played with the halfmove clock at exactly 150: the game ends on the precedence clause of Article 9.6.2. Its length is insensitive to that clause — under either reading the 17,697th ply is played and the game ends on it; the clause decides only whether the result is a win or a draw.
- The profile is the equality case of Theorem 3.8: 117 critical moves plus a closing segment ($K = 118$), with $P = 96$ (each of the sixteen pawns makes exactly six pawn moves), $C = 29$, $O = 8$ (every file resolved), $T = 1$. The endpoint colour sequence has shape BWBW, so $S = 3$, and the decomposition of Lemma 2.4 closes exactly: 115 segments of 150 plies, 3 segments of 149 (one per switch), and $\sum_i \delta_i = 0$:

$$17,697 = 150 \cdot 118 - 3 - 0.$$

Improving on the witness would require eliminating a colour switch rather than packing segments more tightly: it already fills every segment, and its three lost plies are exactly the three boundaries Theorem 4.6 proves unavoidable. The repository also *rebuilds* a different game of the same length — same 118 critical moves, freshly generated quiet filler; 17,032 of the 17,697 plies differ — and verifies it the same way, so attainment does not rest on a single artefact.

7 Verification methodology

The proof above is a hand argument, and we intend it to be checkable entirely by hand. This section records what was additionally checked by machine, and what a sceptical reader must trust. The companion repository [9] packages all of it; Appendix A maps each claim to the code that checks it, every check re-runs from a byte-for-byte-reproducible certificate, and Appendix B analyses the failure mode specific to this style of checking, with the three instances repaired during the work.

7.1 Finite checks

The following case analyses, each also proved in the text, are exhausted mechanically: the unresolved-pair cap of Lemma 3.3, over a state space strictly more permissive than chess (Remark 3.4); the impossibility of two facing pawns passing each other on a file; the uniqueness of the equality assignment of Theorem 3.8; the terminal seam of Lemma 4.1 over every alternating shape of up to twelve blocks; and, for Lemma 4.4, the obligations [H0] (16 squares of the initial position), [H3] (all four castling moves) and [H4] (all 28 en passant configurations), which have finitely many instances and are settled outright.

The remaining obligations [H1], [H2], [H5], [H6] are universally quantified over all positions, so no enumeration can settle them; they are read off the Laws [5], and the repository checks them against a large randomised corpus of positions. That corpus check tests that the move generator we rely on implements those four rules correctly — it is a check on the tooling, not on the mathematics.

In addition, the home-rank lemma was attacked adversarially before it was proved: a search playing real chess under (R1), (R2) — with a control run confirming the search can break through once the restrictions are lifted — examined every legal attacker move at every position visited (4,430,269 moves across 90,779 positions, seeded and reproducible) and found no pawn reaching the defended home rank and no en passant capture ever available. The search is evidence; the induction of Lemma 4.4 is the argument the evidence was for.

7.2 Two independent mechanical decisions of Theorem 4.6

To reduce the risk of an undetected error in the hand argument of Section 4, the question “does any shape with $S \leq 2$ admit $K = 118$?” is re-decided mechanically by two further methods: a CP-SAT constraint model [8], and a solver-free arithmetic checker that declares its own constants and decides the same question by direct enumeration. Their verdicts carry weight only under a precisely stated soundness property, which we give for the constraint model; the arithmetic checker imposes aggregated forms of the same conditions, each likewise a necessary condition of legality.

Fix an alternating block shape σ , an ending $e \in \{\text{checkmate}, \text{draw}\}$, and a target k . The model $M(\sigma, e, k)$ is a finite system of constraints over bounded integer variables: per-pawn pawn-move counts in each block of the pawn’s colour, with a per-pawn total of at most six; per-unit capture indicators, each captured unit assigned a death block of the enemy colour, a captured pawn-origin unit making no pawn move in any later block; capture, overlap and closing-segment totals with the coupling of Proposition 3.6 imposed through a free resolved-file count; the cap of Lemma 4.4 on opening-block pawn moves, imposed only while no enemy pawn has been captured in that block; for $e = \text{checkmate}$, the requirement that the side owning the final block retain at least one non-king unit; and the requirement that the totals compose to k as in (1).

Soundness property. Let G be a legal game with $K(G) = k$ whose endpoint sequence has shape σ , and set e to checkmate if G ends in mate and to draw otherwise. Then the observed values of G — its per-pawn, per-block move counts, its capture blocks, and its totals — satisfy every constraint of $M(\sigma, e, k)$. This holds because each constraint is individually a necessary condition of legality; the constraint-by-constraint argument is

recorded in the repository’s model audit, and the attribution of the mating side to the final block is sound precisely because shapes are sequences of segment endpoints (Definition 2.2), the mate being the last endpoint. Consequently, if $M(\sigma, e, 118)$ is infeasible for both values of e , then no legal game with endpoint shape σ attains $K = 118$. No claim is made in the converse direction: a satisfying assignment need not correspond to any legal game, and no conclusion of this paper rests on a feasibility verdict.

Coverage is finite. An alternating shape of $n \geq 5$ blocks has $S \geq 4$ by the block count of Section 4, so the sixteen decided instances — the eight shapes of at most four blocks, under both endings — include every shape relevant to Theorem 4.6. CP-SAT decides each finite instance exactly; a solver timeout is surfaced as an error and never reported as infeasibility. The results, identical under both endings: every shape with $S \leq 2$ is infeasible, in agreement with the counting proof, and the feasible shapes are exactly BWBW and WBWB. Thus every feasible shape has $S \geq 3$; the converse fails and is not claimed — WBW, with $S = 3$, is also infeasible, a verdict beyond what Theorem 4.6 requires and in no tension with it, since infeasibility only ever removes games. Removing the home-rank constraint changes exactly BWB and WBW from infeasible to feasible and leaves every other verdict unchanged, confirming that the lemma carries precisely the case Proposition 4.5 assigns to it and that no other refutation of Section 4 depends on it.

The soundness property cannot be established by testing, but violations of it can be witnessed: the constraint model accepts the exact assignment induced by the witness game of Section 6, both deciders return feasible for its endpoint shape BWBW, and Appendix B records three constraints that were, at some stage of this work, stronger than legality — each found by the constraint-by-constraint argument, none surviving in the reported verdicts.

7.3 The trusted base

A reader who checks the hand proof of Sections 2–5 and re-runs the repository must still trust the following. (i) The summary of the Laws in Section 2.1, in particular that Articles 9.6.1–9.6.2 end games automatically and that mate takes precedence on the final ply — the witness’s length, though not its result, is insensitive to the precedence clause. (ii) The move generation of `python-chess` [7] (version pinned in the repository), which underlies the witness verification and the corpus checks of Section 7.1; the obligations [H1], [H2], [H5], [H6] are facts about chess that we verify the library respects, not facts we derive from it. The witness is an artefact produced by others [2] and accepted by our verifier, which separates producer from judge; an additional verification under an independently implemented move generator would shrink this trust further and is noted as future work. (iii) That the dead-position rule never shortens the witness: this needs no decision procedure for Article 5.2.2, only the observation of Section 6 that a game ending in mate contains no dead position. The upper bound uses deadness once, for king versus king (Lemma 3.7), where it is beyond dispute; everywhere else, ignoring dead positions only lengthens hypothetical games, which is the safe direction for an upper bound. The counting proof itself is not formalised in a proof assistant; a formalisation (with the human text as its blueprint) is natural future work.

Acknowledgements

This work exists because Tom Murphy VII built a 17,697-ply game and asked, in the title of [2], whether it is the longest; the answer is yes. The structure of that answer — 118 clock-resetting moves, three unavoidable half-move losses — was identified years earlier by François Labelle [1], together with the forum contributors his page credits.

This work was carried out independently, outside any research group and without funding.

The board diagrams were rendered with `python-chess` [7]; the piece images are by Colin M. L. Burnett, triple-licensed under the GFDL, BSD and GPL, and are used here under the BSD option with this attribution.

A Claims-to-repository map

The companion repository [9] verifies the following claims mechanically. The table names each statement and the routine that decides it; module paths, the tests that pin each claim, invocation commands, expected outputs and measured runtimes are catalogued in the repository’s top-level `CLAIMS.md`, and a certificate directory records every verdict together with a hash of every artefact, re-derivable byte for byte from a clean checkout.

Statement	Deciding routine
Lemma 3.3, cap exhausted over a relaxation	<code>check_origin_pair_cap</code>
Lemma 3.3, cap attained legally	the 24-ply legal prefix, replayed
Lemma 3.3, facing pawns cannot pass	<code>check_file_lemma</code>
Proposition 3.6, maximiser unique at $f = 8$	<code>pawn_minus_overlap_bound</code>
Theorem 3.8, equality assignment unique	<code>equality_witnesses</code>
Lemma 4.1, over all shapes of ≤ 12 blocks	<code>check_terminal_endpoint_free</code>
Lemma 4.4, obligations [H0]–[H6]	<code>invariant.verify</code>
Lemma 4.4, adversarial search	<code>adversary.audit</code>
Section 4, the five refutations	<code>switch_lower_bound</code>
Section 7.2, sixteen instances and the ablation	<code>solve, analyse</code>
Section 6, witness verification	<code>verifier</code>
Section 6, the rebuilt second witness	<code>pad_skeleton</code>
Appendix B, regression pins	the repository test suite
certificate reproducibility	<code>certify, check_certificate</code>

The repository version corresponding to this paper is tag `v1.0.0`, cited in [9] together with the archival DOI of that release’s Zenodo deposit.

B Over-constraint: three repaired instances

Impossibility checking over necessary conditions has one failure mode that cannot be detected from its verdicts alone: a constraint stronger than legality produces spurious infeasibility, indistinguishable from a correct refutation. Three such constraints were introduced and later repaired during this work. They are recorded here because the record bears on the auditing standard applied to the surviving constraints, and because none of the three was

found by testing — each was found by the constraint-by-constraint argument of Section 7.2, which is the only mechanism that can find them.

First instance. An early version capped an unresolved origin pair at 4 combined pawn moves — the first half of the proof of Lemma 3.3 mistaken for the whole. The cap is refuted by a legal 15-ply sequence ending with 8. a5, in which a bishop captures one pawn of the pair and the survivor advances freely (Figure 4(a)). The corrected cap of 10 weakens the coupling of Proposition 3.6 at every f while leaving its maximum of 88 unchanged, so Theorem 3.8 was unaffected; the margins by which the five shapes of Section 4 fail had, however, been overstated, and were re-derived.

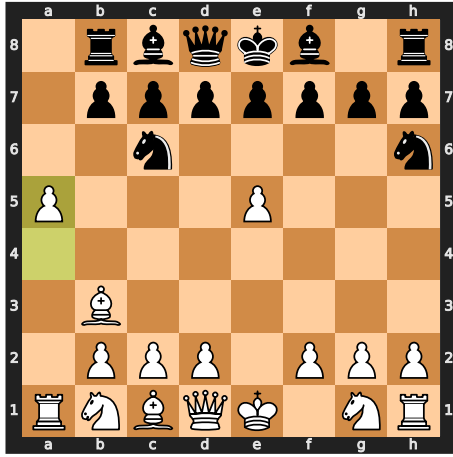
Second instance. The model’s checkmate branch once required the mated side to lose all fifteen of its capturable units. The condition does hold at $K = 118$, where it follows from $C + T = 30$, but it is not a consequence of checkmate — a scholar’s-mate game refutes it — and imposed unconditionally it excluded legal games at other targets. It was weakened to the condition stated in Section 7.2, which is all that checkmate implies.

Third instance. Reading the final block of a shape as the mating side is sound over sequences of segment endpoints and unsound over critical-move sequences alone: a quiet mate can be delivered by the colour that made no critical move last, and an explicit 76-ply legal game realises the gap — its critical actors form (B,W) while Black delivers the mate (Figure 4(b)). The convention had been applied consistently but stated nowhere, and one description of the model used the incorrect reading. The convention is now explicit in the soundness property of Section 7.2, resting on Definition 2.2, and the witnessing game is pinned as a regression.

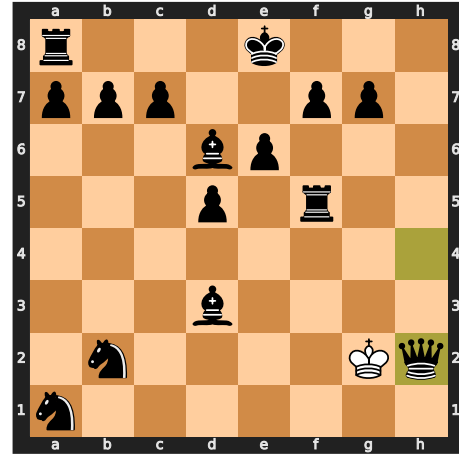
In all three cases the sixteen decided instances were re-derived after repair and no verdict changed. For the second and third instances this is provable in retrospect: the offending constraint is derivable at $K = 118$ (second), and the draw instances, which carry no mating-side condition at all, are infeasible for the same six shapes (third).

References

- [1] F. Labelle. The longest possible chess game, and bounds on the number of possible chess games. Web page, created 14 November 2015, last updated 17 October 2020, accessed August 2026. <https://wismuth.com/chess/longest-game.html>; archived at <https://web.archive.org/web/20260806171843/https://wismuth.com/chess/longest-game.html>.
- [2] T. Murphy VII. Is this the longest Chess game? In *A Record of the Proceedings of SIGBOVIK 2020*, April 2020. <http://tom7.org/chess/longest.pdf>.
- [3] FIDE. *Laws of Chess: for competitions starting from 1 July 2014 till 30 June 2017*, FIDE Handbook E/01. <https://handbook.fide.com/chapter/E012014>.
- [4] FIDE. *Laws of Chess: taking effect from 1 July 2017*, FIDE Handbook E/01. <https://handbook.fide.com/chapter/E012017>.
- [5] FIDE. *FIDE Laws of Chess*, FIDE Handbook E/01, in force from 1 January 2023. <https://handbook.fide.com/chapter/E012023>.



(a)



(b)

Figure 4: The two witnessed instances of Appendix B. (a) First instance, after 8.a5: the a-file’s origin pair has made five combined pawn moves against a cap that asserted four, neither pawn ever leaving the file. (b) Third instance, the final position of the 76-ply game: every Black critical move precedes White’s single one (the king’s capture on g2), and Black has just delivered a quiet mate.

- [6] J. Tromp. The longest chess game. Web note, <https://tromp.github.io/chess/longest.html>, accessed August 2026; archived at <https://web.archive.org/web/20260806171910/https://tromp.github.io/chess/longest.html>.
- [7] N. Fiekas. python-chess: a chess library for Python, version 1.11.2. <https://github.com/niklasf/python-chess>.
- [8] L. Perron and F. Didier. CP-SAT solver, Google OR-Tools. https://developers.google.com/optimization/cp/cp_solver.
- [9] J. Yim. Companion repository for “The maximum length of a chess game under the FIDE Laws is 17,697 plies”. https://github.com/junyeobe0315/long_chess, 2026. Version v1.0.0, archived at <https://doi.org/10.5281/zenodo.21828026>.